ADVANCED SYSTEMS ANALYSIS AND DESIGN

TOPIC 3:

Product Backlogs (PBL) and Agile Estimation System Analysis

CIS 301

Introduction to Agile Methodology

What is Agile?

- Agile is a set of principles and practices for software development that prioritize flexibility, collaboration, and customer satisfaction.
- It's an iterative and incremental approach that allows teams to respond to change and deliver value quickly.

Origins of Agile

- Agile principles were articulated in the Agile Manifesto in 2001 by a group of software developers who sought an alternative to traditional, plan-driven project management.

Introduction to Agile Methodology

Agile Methodologies

- Agile is not a one-size-fits-all approach. Various Agile methodologies exist, such as Scrum, Kanban, Lean, and Extreme Programming (XP).
- Each methodology has its own practices and processes, but they all align with the Agile principles and values.

Iterative and Incremental Development

- Agile promotes iterative development, where a project is broken into small, manageable iterations or sprints.
- Each iteration produces a potentially shippable increment of the product.

Introduction to Agile Methodology

Benefits of Agile

Faster time to market.

Improved quality.

Enhanced customer satisfaction.

Better team morale.

Greater flexibility in responding to change..

Challenges of Agile

Agile is a flexible and customer-focused approach to software development.

It values collaboration, adaptability, and customer satisfaction.

Agile methodologies, like Scrum and Kanban, are widely adopted in the industry for their effectiveness in delivering value quickly.

Agile Framework

What is an Agile Framework?

An Agile framework is a structured approach to implementing Agile principles and practices in a project or organization.

It provides a set of guidelines, roles, and ceremonies to facilitate Agile development

Agile Manifesto Revisited

Reiterate the four core values and twelve principles of the Agile Manifesto.

Explain how these principles guide the development process within Agile framewo

Agile Framework

Scrum Framework

Introduce Scrum as one of the most popular Agile frameworks.

Highlight key elements of Scrum:

Roles (Product Owner, Scrum Master, Development Team).

Artifacts (Product Backlog, Sprint Backlog, Increment).

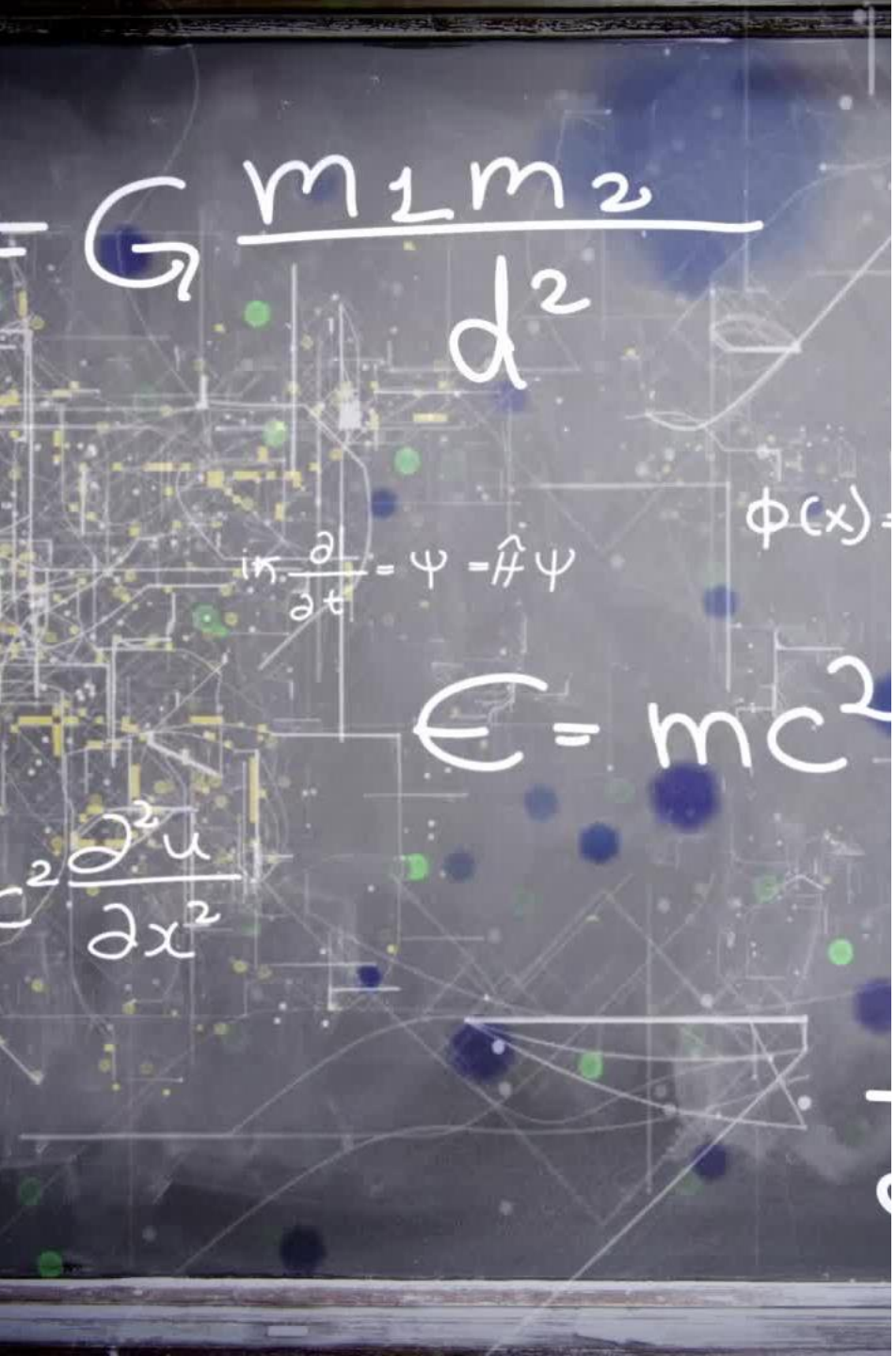
Ceremonies (Sprint Planning, Daily Standup, Sprint Review, Sprint Retrospective).

Kanban Framework

Lean Framework

Scaled Agile Framework (SAFe)

Extreme Programming (XP)



HW2

Explain each of the following in one slide:

- **Kanban Framework**
- **Lean Framework**
- **Scaled Agile Framework (SAFe)** HW presntaion
- **Extreme Programming (XP)**

Product Backlog (PBL)

What is a Product Backlog?

- The Product Backlog (PBL) is a dynamic and prioritized list of features, user stories, and improvements that represent the requirements for a product.

Role of the Product Backlog

- Highlight the importance of the Product Backlog in Agile development.
- Explain that it helps maintain a clear vision, provides transparency, and guides the development team.

Product Backlog (PBL)

Creating a Product Backlog

- Describe the process of creating a Product Backlog:
- Collecting and prioritizing user stories and requirements.
- Involving stakeholders and the product owner.
- Breaking down high-level features into smaller, actionable items
- **Product Backlog Items (PBIs)**

- Explain that items in the Product Backlog are often referred to as Product Backlog Items (PBIs).
- Discuss the various types of PBIs, including user stories, bugs, technical debt, and enhancements.

Creating a Product Backlog

Step 1	Identify Stakeholders
Step 2	Define the Product Vision
Step 3	Gather Requirements
Step 4	Create User Stories
Step 5	Prioritize User Stories
Step 6	Estimate User Stories
Step 7	Create a Definition of Done (DoD)
Step 8	Review and Refine
Step 9	Continuously Add and Remove Items
Step 10	Maintain Transparency

- Creating a Product Backlog is a fundamental step in Agile development.
- It involves identifying stakeholders, defining a clear product vision, gathering and prioritizing requirements, estimating user stories, and maintaining transparency.
- Regular refinement and adaptation are key to a successful Product Backlog

Creating a Product Backlog
Role of the Product Owner

Key Responsibilities

Describe the primary responsibilities of the Product Owner:

- Defining and communicating the product vision.
- Maintaining and prioritizing the Product Backlog.
- Collaborating with stakeholders.
- Making decisions on scope and priorities.
- Ensuring the team delivers value to customers.

User Stories

- Explanation of user stories
- How user stories are used in product backlogs
- Examples of well-defined user stories

User Stories

What is a User Story?

- A user story is a brief, informal description of a feature, told from the perspective of an end-user or customer.
- It outlines what the user wants to achieve and why

Components of a User Story

- **Title**: A short, descriptive name for the story.
- **Narrative**: A description of the story, often following the format "As a [user], I want [feature] so that [benefit]."
- **Acceptance** Criteria: Specific conditions that must be met for the story to be considered complete.

User Stories

Advantages of User Stories

- They focus on the user's perspective and needs.
- They are concise and easy to understand.
- They promote collaboration between team members and stakeholders.

User Stories vs. Traditional Requirements

- Highlight the differences between user stories and traditional, detailed requirements documents.
- Explain how user stories are more flexible and adaptable to change

User Stories

INVEST Criteria are characteristics of well-formed user stories:

- Independent
- Negotiable
- Valuable
- Estimable
- Small
- Testable

How user stories are used in product backlogs

- **User Stories** are a vital element of **Product Backlogs** in Agile development.
- They **capture end-user needs** and describe **desired functionality** from the **user's perspective**.
- **User Stories include**:
  - A brief **title** for easy reference.
  - A **narrative** that describes the user, the desired feature, and the benefit.
  - **Acceptance criteria** detailing specific conditions for completion.
- **Role of User Stories in Product Backlogs**:
  - Serve as the **backbone of requirements**.
  - **Prioritize development work** based on user value.
  - **Enable clear communication** between the team and stakeholders.
  - Promote **flexibility** and **adaptive planning** as user needs evolve.
- **Regularly refined and updated** in collaboration with stakeholders to ensure alignment with the product vision and changing requirements.
- **User Stories empower teams** to deliver valuable increments of the product in an iterative and customer-centric way.

Examples of well-defined user stories

Example 1: Search Functionality

- **Title**: Search Functionality
- **Narrative**: As a website visitor, I want to be able to search for products by keywords to quickly find what I'm looking for.
- **Acceptance Criteria**:
  - Users can enter search terms in a search bar.
  - Search results display relevant products with descriptions.
  - Users can filter search results by category.
  - Clicking on a result item leads to its details page.

Example 2: Password Reset

- **Title**: Password Reset
- **Narrative**: As a forgetful user, I want to reset my password if I can't remember it to regain access to my account.
- **Acceptance Criteria**:
  - Users can request a password reset by providing their email.
  - Users receive a password reset email.
  - Clicking the reset link allows users to set a new password.

Estimation in Agile

Introduction to the concept of estimation in Agile

Why estimation is crucial in Agile development

Different methods of estimation (Story Points, Ideal Days, T-Shirt Sizes)

Estimation in Agile

Estimation in Agile is vital for effective planning and prioritization. is a practice used in Agile methodologies to predict how much work a team can complete within a given timeframe.

Purpose:

- Plan work for a given timeframe.
- Prioritize tasks.
- Facilitate team collaboration.

Key Principles:

- Relative Sizing.
- Collaboration.
- Iterative.
- Practicality.

Common Units:

- Story Points.
- Ideal Days.
- T-Shirt Sizes.

Techniques:

- Planning Poker.
- Fibonacci Sequence.
- T-Shirt Sizing.

Benefits:

- Better planning.
- Resource allocation.
- Improved communication.
- Adaptability to change.

Why estimation is crucial in Agile development

Agile Estimation is **crucial** in Agile development for several reasons:

- Planning and Prioritization
- Resource Allocation
- Predictability
- Adaptability
- Communication and Collaboration
- Alignment with Business Goals
- Continuous Improvement

Different Estimation Methods in Agile

Story Points:

- Relative measure of complexity or effort used to quantify the size of user stories, emphasizing relative comparisons.

Ideal Days:

- A theoretical measure of time required to complete a task, assuming no interruptions or distractions.

T-Shirt Sizes:

- A simplified way to represent task size, often categorized as Small, Medium, Large, etc., to avoid precise time or effort estimates.

Agile Estimation Techniques

Planning Poker: Collaborative method where team members discuss and vote on the effort or complexity of user stories using cards.

Fibonacci Sequence: A sequence of numbers (e.g., 1, 2, 3, 5, 8, 13, ...) used for relative sizing to avoid assigning exact hours or days.

T-Shirt Sizing: Assigning relative sizes (e.g., S, M, L, XL) to tasks to quickly convey their complexity without precise time estimates.

Estimation Accuracy

Definition: The degree to which estimated work matches actual effort and completion time.

Factors Influencing Accuracy:

Team Experience: More experienced teams tend to have better estimation accuracy.

Complexity: Complex tasks are often less accurately estimated.

Scope Changes: Frequent changes to project scope can impact accuracy.

Communication: Effective team communication improves accuracy.

Improving Accuracy:

Regularly Review and Refine Estimates.

Learn from Past Projects.

Involve the Entire Team in Estimation.

Use Historical Data.

Importance:

Accurate estimates enhance project planning and decision-making.

Helps in meeting stakeholder expectations.

Drives a culture of continuous improvement.

Important Terms

Velocity:

- *Definition*: The measure of a team's capacity to complete work in a sprint, typically expressed in story points or other units.
- *Purpose*: Helps teams plan future sprints, set realistic goals, and gauge their productivity.
- *Calculation*: Sum of story points completed in a sprint over multiple sprints.

Release Planning:

- *Definition*: A strategic Agile practice that involves long-term planning and prioritization of features to be delivered in multiple sprints.
- *Purpose*: Helps align development efforts with business goals and customer needs.
- *Outcome*: A roadmap outlining which user stories or features will be delivered in each release.

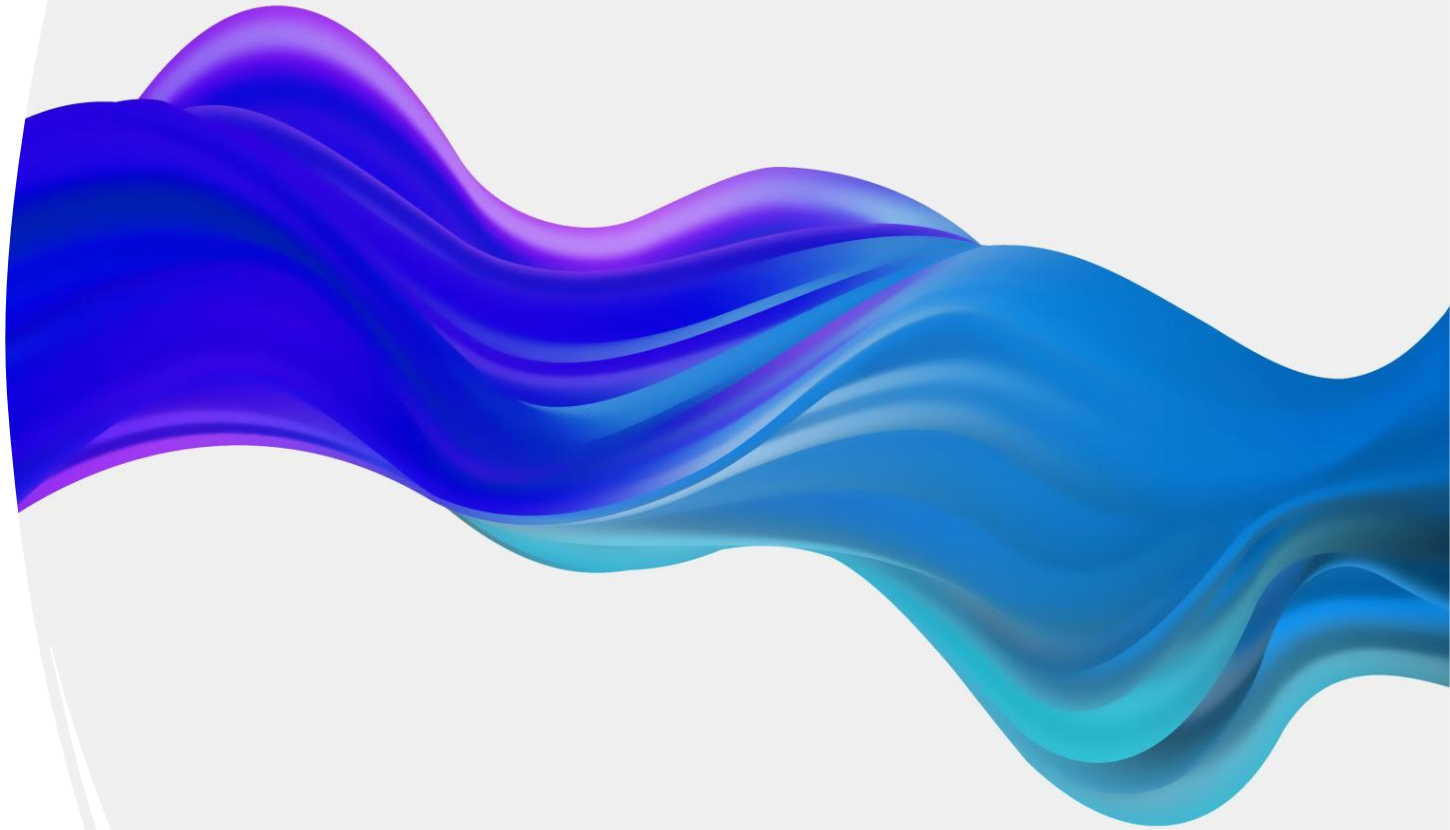
Monitoring and Adjusting:

- *Definition*: Ongoing tracking and evaluation of project progress, with the flexibility to make necessary changes.
- *Purpose*: Ensures that the project stays on course, adapts to changes, and meets evolving customer needs.
- *Activities*: Regularly reviewing sprint progress, adjusting priorities, and making data-driven decisions.

Common Tools for Agile Estimation

1. **Jira**: A widely used project management tool that offers Agile-specific features like backlog management, sprint planning, and velocity tracking.
2. **Trello**: A visual collaboration tool with Kanban boards that teams use for Agile project management and estimation.
3. **Azure DevOps**: Microsoft's toolset for planning, developing, testing, and delivering software, including Agile estimation and tracking features.
4. **VersionOne**: An Agile management tool known for its support of Scrum, Kanban, and SAFe methodologies, including estimation capabilities.
5. **Poker Planning Tools**
6. **Agile Board in GitLab**
7. **Targetprocess**
8. **Monday.com**
9. **Smartsheet**
10. **Custom Spreadsheets**

Best Practices

- **Prioritization**: Prioritize backlog items based on business value, customer needs, and strategic goals.
- **Clear User Stories**: Ensure user stories are well-defined, concise, and aligned with user needs.
- **Continuous Refinement**: Regularly review and refine the backlog to keep it up-to-date and ensure items are ready for development.
- **Acceptance Criteria**: Define clear and testable acceptance criteria for user stories.
- **INVEST Criteria**: Follow the INVEST criteria (Independent, Negotiable, Valuable, Estimable, Small, Testable) for creating effective user stories.
- **Slicing User Stories**: Break down large user stories into smaller, valuable increments for easier estimation and development.
- **Collaboration**: Promote collaboration between the product owner, development team, and stakeholders for shared understanding.
- **Estimation Consistency**: Maintain consistency in estimation units (e.g., story points) for accurate comparisons.

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